

Arif Solmaz, PhD

Assistant Professor — Exoplanet Science & Scientific ML

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PROFILE

Astronomer specializing in exoplanet transit photometry and timing analysis, with emphasis on characterizing systematic effects from stellar activity. I develop Bayesian and machine learning pipelines to extract precise transit parameters from TESS, JWST, and ground-based data, specializing in transit timing variations (TTVs) and starspot-induced biases. As a lead contributor to international collaborations including the ExoClock Project and EXPLORE, I coordinate multi-site observing campaigns across 50+ countries. I teach computational methods and programming at the undergraduate level and maintain reproducible, open-source research workflows.

EDUCATION

PhD in Physics

Çukurova University

Jan 2023

Adana, Turkey

- Dissertation: *Effects of Starspot Occultations on Exoplanet Transit Mid-time Measurements*
- Advisors: Prof. Dr. Aysun Akyüz & Prof. Dr. Özgür Baştürk (Ankara University)

MSc in Space Sciences and Technologies

Çanakkale Onsekiz Mart University

Oct 2010

Çanakkale, Turkey

- Thesis: *Extrasolar Planetary Systems: a status review*
- Advisor: Prof. Dr. Mehmet Emin Özel

BSc in Physics

Çanakkale Onsekiz Mart University

Jan 2008

Çanakkale, Turkey

- Advisor: Prof. Dr. Faruk Soyduğan

ACADEMIC APPOINTMENTS

Assistant Professor

İstanbul Health and Technology University (İSTÜN)

May 2024 – Present

Istanbul, Turkey

Department of Mechatronics Engineering, Faculty of Engineering & Natural Sciences. Teaching programming, data structures, and machine learning courses; leading research in exoplanet science and applied ML.

Lecturer

Çağ University, Faculty of Arts and Sciences

Sep 2015 – Jun 2023

Mersin, Turkey

Taught statistics, programming, and quantitative methods while completing PhD in parallel. Developed digital learning materials and supervised student projects.

Research Assistant

Çağ University, Faculty of Arts and Sciences

Feb 2011 – Sep 2015

Mersin, Turkey

Supported teaching and observational research activities; contributed to TÜBİTAK-funded meteor monitoring and stellar astrophysics programs.

SELECTED PUBLICATIONS

Complete list at arifsolmaz.github.io.

White Papers / Community Papers:

- The role of the Hubble Space Telescope in advancing our understanding of atmospheric escape in exoplanets. *arXiv e-prints*, arXiv:2605.26140, 2026 · STScI Hubble 2030s roadmap white paper · co-author

Peer-Reviewed Journal Articles:

- A high geometric albedo and small size for the Haumea cluster member (24835) 1995 SM55 determined from a stellar occultation and photometric observations. *Astronomy & Astrophysics*, 703, A147, 2025
- Testing the performance of cross-correlation techniques to search for molecular features in JWST NIRSpec G395H observations of transiting exoplanets. *MNRAS*, 543, 3456, 2025
- ExoClock Project IV: A homogeneous catalogue of 620 updated exoplanet ephemerides. *ApJS*, 283, 5, 2026 · 100+ authors
- ExoClock Project III: 450 New Exoplanet Ephemerides from Ground and Space Observations. *ApJS*, 265, 4, 2023 · 60+ authors
- Leke Örtümlerinin Ötegezegen Geçiş Ortası Zaman Ölçümlerine Etkisi. *Turkish Journal of Astronomy and Astrophysics*, 4, 147, 2023
- Physical properties of the trans-Neptunian object (38628) Huya from a multi-chord stellar occultation. *A&A*, 664, A130, 2022
- BO Ari Light Curve Analysis using Ground-Based and TESS Data. *New Astronomy*, 86, 101571, 2021
- The First Light Curve Solutions and Period Study of BQ Ari. *Astronomy Letters*, 47, 402, 2021

Selected Conference Contributions:

- Artificial Intelligence in Planetary Science and Astronomy: Applications and Research Potential. EPSC-DPS Joint Meeting 2025 · Invited presentation
- Europlanet Machine Learning Working Group: a year of progress. EPSC-DPS Joint Meeting 2025

FUNDED PROJECTS

As Principal Investigator / Coordinator:

- **Geometric Modeling and Analysis of Exoplanet Transit Observations** — University Scientific Research Project. 2019–2020.
Bayesian parameter estimation from transit photometry; handling stellar activity systematics.
- **MILAGE: Technology Use in Mathematics Education** — EU Erasmus+ Programme. 2015–2018.
Coordinated multi-country partnership on digital tools in education.
- **Turkey Meteor Monitoring Systems and Network (Continuation)** — University Scientific Research Project. 2019–2020.
- **Light Pollution Measurement Studies** — University Scientific Research Project. 2019–2020.

As Team Member:

- **The Impact of Non-Markovian Memory Effects on the Learning Dynamics of Quantum Neural Networks** — TÜBİTAK. *Submitted / Under Review*.
Investigation of non-Markovianity and memory effects in the optimization and training trajectories of quantum machine learning models.
- **ML-Based Microscopic Wood Species Identification System** — TÜBİTAK-1002. *Ongoing*.
Deep learning models for automated classification from microscopic images.
- **Turkey Meteor Monitoring Systems and Network: National Impact Craters and Meteorites Database** — TÜBİTAK-1001. 2014–2017.
- **Multi-faceted Analyses of Binary and Multiple Star Systems** — TÜBİTAK-1010 EVRENA. 2010–2013.

SOFTWARE & TOOLS

- **TransitKit**: Python toolkit for exoplanet transit light curve analysis — Bayesian fitting, MCMC sampling, systematic detrending, publication-quality output
- **arXiv Monitor**: Automated daily aggregator for exoplanet literature with AI-generated summaries ([source](#))

TECHNICAL SKILLS

- **Programming:** Python, C, Bash; Git/GitHub; Linux environments
- **Scientific Stack:** NumPy, SciPy, Pandas, Matplotlib; Astropy, Photutils, Lightkurve
- **Statistics & ML:** Bayesian modeling, MCMC (emcee, PyMC), Gaussian processes (celerite); Scikit-learn, TensorFlow/Keras; CNNs, transfer learning
- **Other:** LaTeX, Jupyter; Turkish (native), English (advanced — YÖKDİL 87.5/100)

COLLABORATION & SERVICE

- **ExoClock Collaboration:** Ground-based observation coordination and ephemeris refinement; 100+ observers, 50+ countries; supporting ESA Ariel mission scheduling
- **EXPLORE Project:** International TTV observing campaign; contributing photometric data and analysis to the AAVSO Exoplanet Database pipeline
- **Europlanet ML Working Group:** Coordinating community efforts in AI/ML applications for planetary science
- **Multi-Site Stellar Occultation Campaigns:** Data analysis and coordination for physical characterization of trans-Neptunian objects
- **Peer Review:** Referee for international astronomy and planetary science journals

TEACHING

Current (2026–2027, İSTÜN): Physics I & II, Computer Programming I & II, Object-Oriented Programming, Data Structures & Algorithms, Robotics, Machine Learning — all in English and/or Turkish.

Previous (Çağ University): Statistics, Quantitative Methods, Introductory–Advanced Programming.

OUTREACH & LEADERSHIP

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| • ESO Science Outreach Network , Turkey Contact Point | 2010–2022 |
| • Universe Awareness (UNAWA) , Turkey Representative | 2011–2015 |
| • BAP Coordinator , Çağ University Research Projects Unit | 2020–2021 |
| • Executive Board , Çağ University Observatory (UZAYMER) | 2012–2023 |
| • Multiple TÜBİTAK-funded science communication projects | 2018–2026 |

PROFESSIONAL MEMBERSHIPS

International Astronomical Union (IAU), European Astronomical Society (EAS), Turkish Astronomical Society (TAD), European Association for Astronomy Education (EAAE — Board Member).

Complete publication list, research code, and teaching materials at arifsolmaz.github.io. References available upon request.